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Social Determinants of Health and Neurobiology Across the Schizophrenia Course
A Systematic Review

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Supplemental content

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IMPORTANCE Individuals with schizophrenia-spectrum psychotic conditions (SSPCs) are disproportionately exposed to adverse social determinants of health (SDOHs). These exposures drive health inequity in schizophrenia and are consistently linked to earlier illness onset, greater symptom severity, and poorer long-term outcomes. The brain-based mechanisms through which structural disadvantage becomes biologically embedded are not well characterized.

OBJECTIVE To examine structural, functional, neurochemical, and plasticity brain changes associated with SDOHs with, or at risk for, SSPCs.

EVIDENCE REVIEW PsycInfo, PubMed, and Web of Science databases were searched from database inception until January 2026 to identify original, empirical studies that reported on the association between SDOHs (early life adversity, social disconnection, racism/discrimination, poverty, food insecurity) and neurobiological measures (brain structure, brain function, neurochemical, neuroplasticity) in individuals across the schizophrenia spectrum. Quality assessment scores using the Joanna Briggs Institute Checklist weighed the impact of individual study findings on the overall outcome summary.

FINDINGS The systematic literature search yielded 14 500 articles, with 114 articles meeting full inclusion criteria. Most articles examined early life adversity (n = 95) with the remaining articles focused on social disconnection (n = 13), racism/discrimination (n = 4), poverty (n = 2), and food insecurity (n = 1). Regarding brain measures, 83 articles reported on structural brain measures, 23 on functional brain measures, 10 on neurochemical measures, and none on neuroplasticity. Together, these studies recruited 10 921 participants across the schizophrenia spectrum (high schizotypy n = 250; familial high-risk n = 1825; clinical high-risk for psychosis n = 3538; first-episode psychosis n = 1169; recent-onset psychosis n = 506; chronic schizophrenia n = 3754). Strong evidence that greater exposure to adverse SDOHs may be associated with neurobiological abnormalities, including cortical thinning, regional brain volume reduction, decreased structural connectivity and functional activation, and abnormal neurochemical levels was found.

CONCLUSIONS AND RELEVANCE These findings support the hypothesis that social adversity becomes biologically embedded, providing evidence for an intermediary link between SDOHs and clinical outcomes in SSPC. Integrating neuroscience with social epidemiology can advance psychosis prevention, refine treatment, and address structural drivers of mental health disparities.

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Social and environmental factors are key drivers of health and health equity.[1,2] The World Health Organization defines social determinants of health (SDOHs) as the conditions in which individuals are born, grow, live, work, and age—shaped by the distribution of power, resources, and opportunity.[3] These factors account for 30% to 55% of health outcomes, outweighing the effects of other medical factors.[1,4]

Individuals with schizophrenia-spectrum psychotic conditions (SSPCs) are disproportionately exposed to adverse SDOHs,[5,6] including early life adversity, social disconnection, racism/discrimination, poverty, and food insecurity. These exposures are linked to earlier illness onset, greater symptom severity, and poorer long-term outcomes,[5,7-13] prompting interest in equity-focused, SDOH-targeted interventions.[2,14,15] It is posited that SDOH exposure may contribute to the development and exacerbation of brain abnormalities,[16] abnormalities consistently associated with worse clinical trajectories in SSPC.[17-24] Yet, the brain-based mechanisms through which structural disadvantage becomes biologically embedded are unknown.

Building on the conceptual framework proposed in a recent trilogy of reviews,[5,16,25] this systematic review provides, to our knowledge, the first empirical synthesis examining how SDOHs are associated with neurobiological outcomes across the schizophrenia spectrum. While prior models outlined theoretical pathways (SDOH exposure→brain abnormalities→poor clinical outcomes),[16] few studies evaluated brain-based mechanisms. Identifying the biological mechanisms impacted by SDOH exposure in psychosis can inform the development of targeted, equitable care.

To address this critical gap, we reviewed studies examining structural, functional, neurochemical, and neuroplasticity brain changes associated with SDOHs in individuals with, or at risk for, SSPCs. In doing so, we clarify how SDOHs get under the skin and shape brain biology.

Key Points

Question Social determinants of health (SDOHs) are linked to worse clinical outcomes in schizophrenia-spectrum conditions; yet, how do SDOHs impact neurobiology across the course of schizophrenia?

Findings In this systematic review, greater exposure to adverse SDOHs was associated with neurobiological abnormalities in schizophrenia, including cortical thinning, brain volume reduction, decreased structural connectivity and functional activation, and abnormal neurochemical levels.

Meaning These findings provide mechanistic support that social adversity becomes biologically embedded, linking SDOHs to clinical outcomes in schizophrenia and underscoring the importance of equity-oriented approaches that integrate social and biological determinants.

Methods

Article Search

Following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guidelines, this review was preregistered on PROSPERO (CRD42024608435). We used PsycInfo, PubMed, and Web of Science to identify published empirical articles from research inception to January 2026, excluding reviews and meta-analyses, that reported on the association between SDOH and neurobiological measures in individuals across the schizophrenia spectrum. We systematically searched using the Boolean search string detailed in the eMethods in the Supplement.

Screening Process and Data Extraction

Article abstracts identified by PsycInfo, PubMed, and Web of Science were independently screened by both authors (J.P.Y.H. and K.D.B.). Both authors independently did a full-text review on articles that passed the screening level. Disagreement at abstract screening and full-text review were discussed until consensus was met.

Data were independently extracted by both authors, with discrepancies resolved through consensus. Extracted study data included (details in the eMethods in the Supplement) author, year, journal, sample size and participant/group characteristics, antipsychotic medication status, type/description of SDOH, type/description of neurobiological measure, neurobiological measurement parameters, confounding factors, and analysis metrics (dataset available online[26]).

Quality Assessment

The Joanna Briggs Institute Checklist for Analytical Cross Sectional Studies[27] (details in the eMethods in the Supplement) was used to examine the methodological rigor of included studies. An additional item was added requiring that studies have an SSPC sample size of 30 or more. Quality assessment scores were used to weigh the impact of individual study findings on the overall outcome summary, with more rigorously sound studies taking precedence.

Results

Study and Participant Characteristics

The literature search yielded 14 500 articles (eFigure 1 in the Supplement). There were 244 articles that were identified for full-text review and 114 articles were included. A total of 97 were case-control studies and 17 studies focused solely on SSPC. Studies were published between 1984 and 2025, with increasing number of studies in the past 15 years (eFigure 2 in the Supplement).

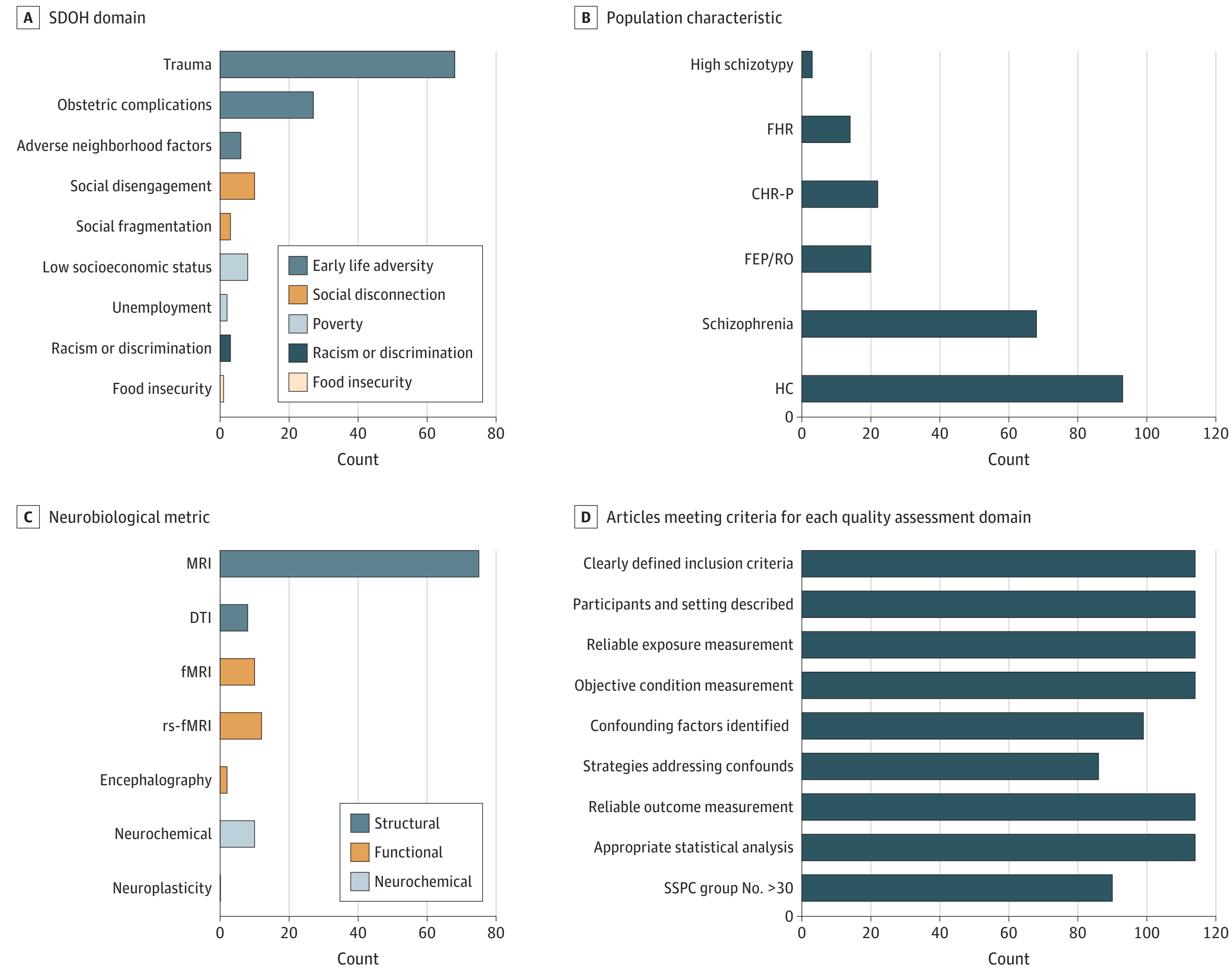
SDOH Characteristics

Most articles focused on early life adversity (n = 95), with emphasis on trauma (n = 68), obstetric complications (n = 27), and adverse neighborhood factors (n = 6) (**Figure 1**A). Social disconnection was examined in 13 articles, including 10 on social disengagement and 3 on social fragmentation. Four articles addressed racism/discrimination. There were 10 articles that explored poverty-related factors, with 8 addressing poverty and 2 unemployment. One article examined food insecurity.

Population Characteristics

Most articles focused on chronic schizophrenia (Figure 1B) (68 articles; n = 3754). Other populations examined were high schizotypy (those scoring high on unusual thinking/odd beliefs; 3 articles; n = 250), familial high risk (FHR) (those at genetic high risk; 14 articles; n = 1825), clinical high risk for psychosis (CHR-P) (22 articles;

Figure 1. Bar Graphs Showing Article Characteristics



CHR-P indicates clinical high risk for psychosis; DTI, diffusion tensor imaging; FHR, familial high risk, FEP/RO, first-episode psychosis/recent onset; HC, healthy controls; fMRI, functional magnetic resonance imaging; MRI, magnetic resonance imaging; rs-fMRI, resting-state functional magnetic resonance imaging; SDOH, social determinants of health; SSPC, schizophrenia-spectrum psychosis conditions.

n = 3538), first-episode psychosis (FEP) (16 articles; n = 1169), and recent-onset (RO) psychosis (4 articles; n = 506).

Neurobiological Characteristics

There were 82 studies that reported on structural brain measures, 23 on functional brain measures, 10 on neurochemical measures, and 0 on neuroplasticity (Figure 1C).

Quality Assessment

Scores ranged between 6 and 9 (eTables 1, 3, 5, 7, and 9 in the Supplement). Confounding factors were identified in 99 of 114 studies (Figure 1D) and 13 of 99 studies did not fully address confounds. While age and sex were commonly addressed, most did not adequately account for antipsychotic status/dosage.

Early Life Adversities and Neurobiological Outcomes

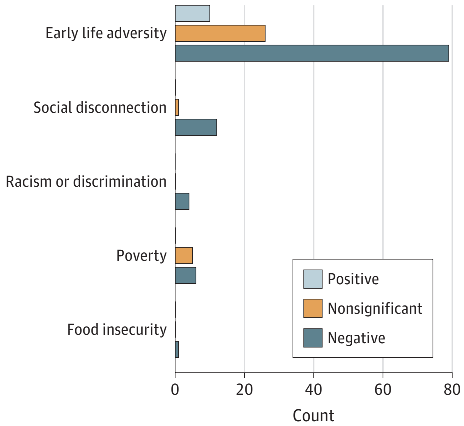
Trauma

There were 68 studies that examined childhood (60 studies)/lifetime (8 studies) trauma effects on the brain, with the majority finding detrimental associations (**Figure 2**A-C; **Figure 3**; **Figure 4**A; eTables 1 and 2 in the Supplement). There were 27 articles that reported greater detrimental associations in SSPC vs healthy controls (HCs), whereas 19 found equivalent group associations (Figure 2D). Most measures focused on severity (57 studies), with some on exposure (8 studies) or frequency (3 studies).

Greater childhood/lifetime trauma was associated with structural (cortical thinning, volume reduction [particularly amygdala and hippocampus], and decreased structural connectivity) and functional brain abnormalities in schizophrenia (33 of 40 studies) and CHR-P (10 of 11 studies), and decreased brain-derived neurotrophic factor and volume reduction (amygdala and orbitofrontal cortex) in FEP/RO (14 of 14 studies) (Figure 3). Greater trauma was associated with lower n-acetylaspartate and greater lactate and glutamate in chronic schizophrenia (2 of 2 studies) and elevated striatal dopamine function in CHR-P (1 of 1 study). While trauma as a whole (54 studies) was often examined, 29 studies looked at subtypes. Greater sexual abuse (10 studies) and

Figure 2. Bar Graphs and Charts Showing a Summary of the Results

A Findings collapsed across neurobiological metrics and SSPCs



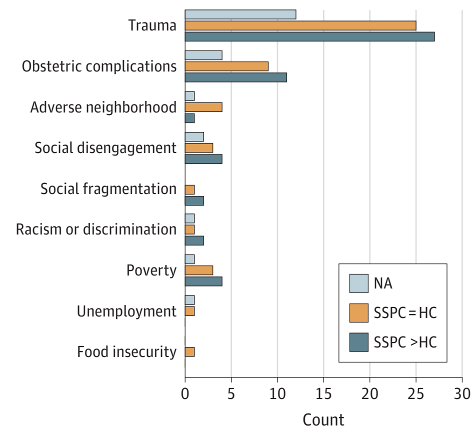
B No. of articles finding detrimental associations between SDOH and neurobiological domains across SSPCs

Subdomain	Structural	Functional	Neurochemical	Neuroplasticity
Trauma	38	15	7	-
Obstetric complications	17	-	1	-
Adverse neighborhood	1	1	-	-
Social disengagement	8	1	-	-
Social fragmentation	3	-	-	-
Racism or discrimination	2	-	2	-
Poverty	4	-	-	-
Unemployment	2	-	-	-
Food insecurity	1	-	-	-

C Percentage of articles finding detrimental associations between SDOH and neurobiological domains across SSPCs

Subdomain	Structural	Functional	Neurochemical	Neuroplasticity
Trauma	86%	83%	100%	-
Obstetric complications	65%	-	100%	-
Adverse neighborhood	33%	33%	-	-
Social disengagement	100%	50%	-	-
Social fragmentation	100%	-	-	-
Racism or discrimination	100%	-	100%	-
Poverty	57%	-	-	-
Unemployment	100%	-	-	-
Food insecurity	100%	-	-	-

D Direction of case-control findings across neurobiological metrics and SSPCs



HC indicates healthy controls; NA, not applicable; SDOH, social determinants of health; SSPC, schizophrenia-spectrum psychosis conditions.

physical abuse/punishment (9 studies) showed the strongest associations, with these subtypes associated with decreased volume and functional connectivity abnormalities of the amygdala and limbic networks. A handful of studies found trauma was either not significantly associated, or positively associated, with brain abnormalities in chronic schizophrenia. Multiple of these studies had small samples or reported both positive/negative associations.

Findings were mixed regarding associations between childhood/lifetime trauma and the brain for FHR and high schizotypy. Across these 11 studies (8 case-control studies), 3 reported a negative association, 2 reported a positive association, and 6 reported no association.

Obstetric Complications and Perinatal Problems

There were 27 studies that examined obstetric complication effects on the brain (Figure 2A-C; Figure 4A; eTables 1 and 2 in the Supplement), with the majority finding detrimental associations (18 studies). There were 11 studies that reported greater detrimental associations in SSPC vs HC and 2 found equivalent group associations (Figure 2D). A total of 17 studies found that obstetric complications, such as general complications (hypoxia, asphyxia) and prenatal infection, were associated with structural brain abnormalities, particularly larger ventricles and decreased hippocampal volumes/integrity, across the schizophrenia illness course. One study found birth hypoxia was associated with decreased brain-derived neurotrophic factor in chronic schizophrenia but not HC.[28] Although 8 found no associations, most involved small samples or drew from overlapping cohorts.

Adverse Neighborhood Factors

There were 6 studies that examined area-level geocoded adverse neighborhood factor effects (Figure 2; Figure 4A; eTables 1 and 2 in the Supplement). One study found greater urbanicity was associated with abnormal amygdala activation in CHR-P and FEP relative to HC.[29] Another reported greater air pollution (PM2.5) was associated with decreased hippocampal volume integrity at baseline and 52-week follow-up in FEP.[30] Four did not find adverse neighborhood factors to be associated with brain volume or resting-state connectivity in SSPC or HC groups.

Figure 3. Charts Showing a Summary of Results by Schizophrenia-Spectrum Psychosis Condition (SSPC) Population

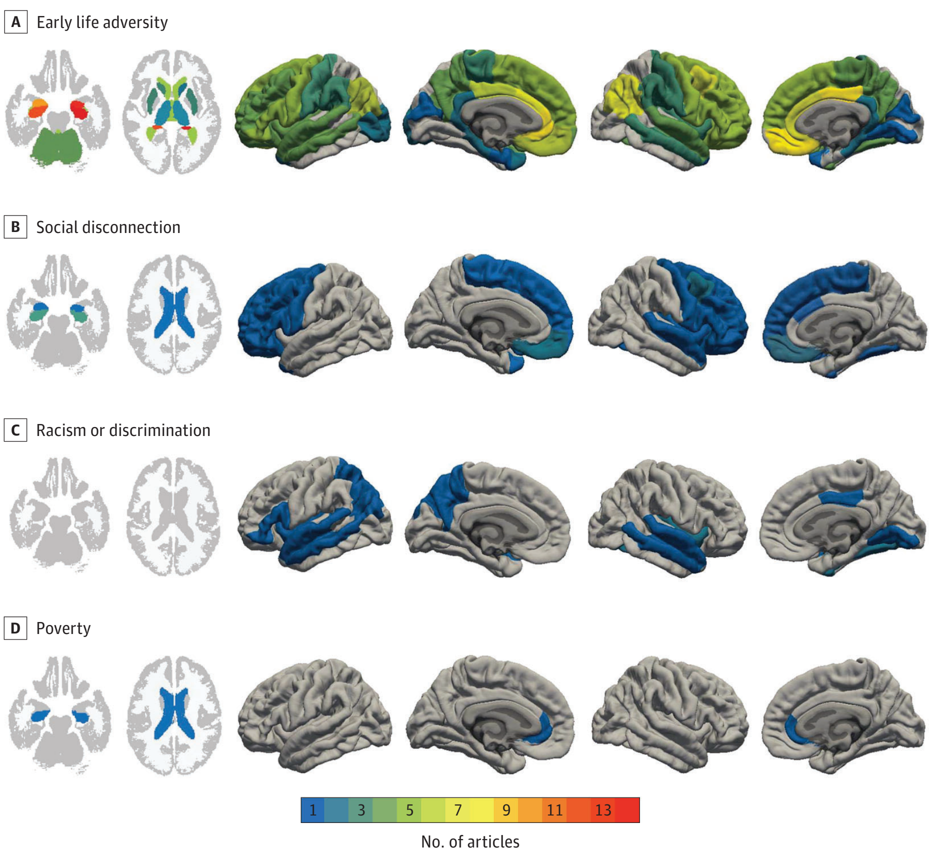
A No. of articles finding detrimental trauma associations

SSPC population	Structural	Functional	Neurochemical	Neuroplasticity
High schizotypy	1	-	-	-
Familial high risk	5	-	-	-
Clinical high risk	8	3	1	-
First episode or recent onset	9	1	4	-
Chronic schizophrenia	22	12	2	-

B Percentage of articles finding detrimental trauma associations

SSPC population	Structural	Functional	Neurochemical	Neuroplasticity
High schizotypy	50%	-	-	-
Familial high risk	71%	-	-	-
Clinical high risk	89%	100%	100%	-
First episode or recent onset	100%	100%	100%	-
Chronic schizophrenia	85%	80%	100%	-

Figure 4. Images Showing Brain Regions Associated With Social Determinants of Health From Structural Morphometric Magnetic Resonance Imaging and Functional Magnetic Resonance Articles



Note that we mapped brain regions across articles to their corresponding brain regions on FreeSurfer's Desikan-Killiany and subcortical atlases. Results implicating global brain deficits are not graphically represented in this figure and are listed in eTables 2, 4, 6, and 8 in the Supplement.

Social Disconnection and Neurobiological Outcomes

Social Disconnection

Ten studies examined social disconnection effects on the brain (Figure 2A-D; Figure 4B, eTables 3 and 4 in the Supplement), with all but one[31] finding detrimental associations. Greater bullying and social distress/anxiety were associated with smaller brain volumes, particularly the hippocampus, in early stages of the schizophrenia course, from high schizotypy to RO. Social disengagement was associated with smaller hippocampal volumes and decreased thickness in CHR-P. Social network deficits were associated with larger ventricles in RO and lower social network brain activity in chronic schizophrenia but not HC.

Social Fragmentation

Three studies examined social fragmentation effects on the brain (Figure 2; Figure 4B; eTables 3 and 4 in the Supplement), with all finding detrimental associations. Greater area residential instability (neighborhood characteristic used to measure social fragmentation[32]) was associated with reduced middle frontal gyrus and cingulate volumes in CHR-P and HC[33] and greater electroencephalogram deficits in CHR-P but not HC.[34] Greater social fragmentation was associated with hippocampus-temporal pole hyperconnectivity in CHR-P, in youth with low/mean, but not high, social engagement levels.[35]

Racism/Discrimination and Neurobiological Outcomes

Four studies examined racism/discrimination effects on the brain (Figure 2; Figure 4C; eTables 5 and 6 in the Supplement), with all finding detrimental associations. Greater racism/discrimination was associated with less brain-derived neurotrophic factor in FEP[36] and widespread decreased thickness in HC and CHR-P.[37] Lower ethnoracial density, which has been linked to greater levels of discrimination,[38] was associated with smaller fusiform gyrus and insula thickness in CHR-P relative to HC, in youth with low, but not high, social engagement levels.[39] Additionally in HC, CHR-P, and chronic schizophrenia, elevated striatal dopamine and synthesis capacity were present in immigrants vs nonimmigrants.[40]

Poverty and Neurobiological Outcomes

Poverty

Eight studies examined poverty effects on the brain (Figure 2; Figure 4D; eTables 7 and 8 in the Supplement), with half finding detrimental associations. Lower family socioeconomic status and greater poverty were associated with smaller regional volumes and decreased thickness in chronic schizophrenia. One study found neighborhood poverty was associated with smaller hippocampal volumes in CHR-P, in youth with low, but not high, social engagement levels.[41] A case-control study did not observe this relationship in chronic schizophrenia.[42] Two small sample case-control studies found socioeconomic status was not associated with cortical thickness in chronic schizophrenia[43] or glutamate levels in FHR.[44] Additionally, a large schizotypy study found socioeconomic status was not associated with brain structure.[45]

Unemployment

Two studies[46,47] examined unemployment effects on brain structure (Figure 2; Figure 4D; eTables 7 and 8 in the Supplement) and found detrimental associations with unemployment, but not homelessness. In HC and chronic schizophrenia, there was a greater ventricle-to-brain ratio in unemployed vs employed individuals.[46] Greater intracerebrospinal volumes predicted unemployment but not homelessness in RO.[47]

Food Insecurity and Neurobiological Outcomes

One small sample study[48] examined prenatal famine exposure on brain structure (Figure 2; eTable 9 in the Supplement). Intracranial volume was decreased in chronic schizophrenia with famine exposure vs without. In HC and chronic schizophrenia, famine exposure was associated with more white matter hyperintensities.

Confounding Factors and Moderators

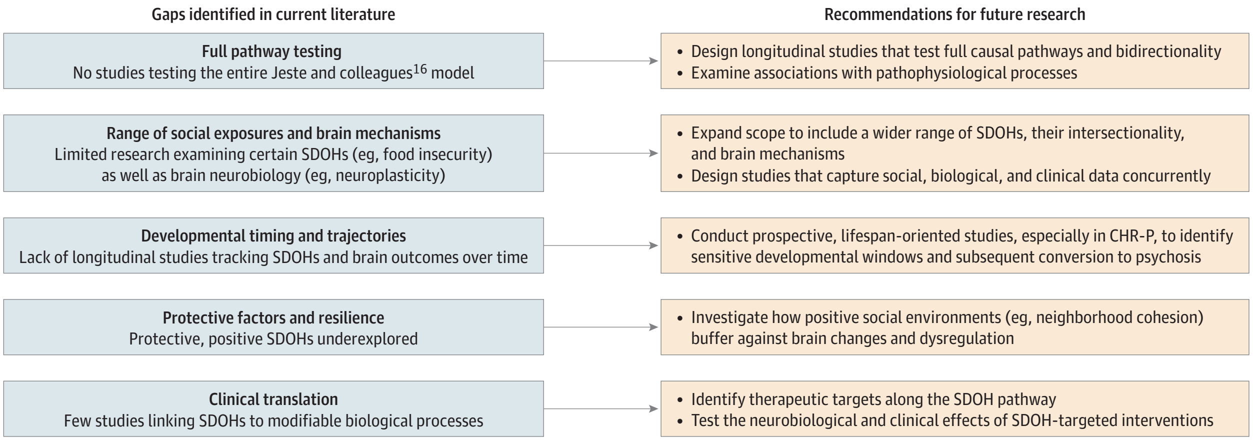
Studies examined a variety of confounding factors, including age, sex, intracranial volume, illness duration/onset age, and antipsychotic medication status/dosage, with little evidence that these factors explained associations between SDOHs and brain abnormalities. Only a few examined secondary SDOHs (eg, social engagement) as a potential moderator, with social engagement moderating associations between SDOHs and brain abnormalities.[35,39,41]

Discussion

Bridging conceptual models with empirical evidence is a priority. Our findings provide empirical support for this theoretical framework[16] by demonstrating that greater SDOH exposure is predominantly associated with detrimental decreases in cortical thickness and volume, increases in striatal and ventricular volume, decreases in structural and functional connectivity, and abnormal neurochemical levels.

The strongest and most extensively examined area implicated trauma and obstetric complications. Multiple studies identified structural and functional abnormalities of the amygdala and hippocampus, key targets of the hypothalamic-pituitary-adrenal (HPA) axis,[49] the brain's stress response system. Schizophrenia is characterized by amygdala and hippocampal abnormalities[18,50,51] and a blunted cortisol awakening response,[52] both closely linked to HPA dysfunction.[49,53] One influential hypothesis proposes that blunted cortisol awakening responses represent an adaptive downregulation of the HPA axis following sustained chronic stress-related hyperreactivity.[54] Consistent with this, most case-control studies of early life adversity reported more pronounced brain abnormalities in SSPC than HC. This pattern suggests heightened sensitivity to adverse SDOHs among SSPC, potentially reflecting combined effects of reduced amygdala and hippocampal volumes and HPA axis dysregulation. Given that SSPC individuals report more frequent and severe SDOHs,[5,55] greater neurobiological abnormalities may reflect greater severity. Notably, among the few studies that matched groups on SDOH severity, adverse effects remained more pronounced in SSPC, consistent with long-standing evidence that individuals with SSPC exhibit heightened trauma sensitivity.[55,56] There was modest evidence for detrimental associations between social disconnection and neurobiological abnormalities, particularly the hippocampus. All 3 studies[33-35] examining social fragmentation reported a negative association, suggesting that social fragmentation may be particularly detrimental. Evidence regarding racism/discrimination was limited, with all showing a negative association.

Figure 5. Chart Showing Recommendations for Future Research



CHR-P indicates clinical high risk for psychosis; SDOH, social determinants of health.

A subset of studies reported nonsignificant or positive associations between SDOHs and the brain. Many of these were in high schizotypy, FHR, CHR-P, and RO groups, suggesting that associations are not as strong early in the schizophrenia course. Certain domains, such as adverse neighborhood factors and poverty, show weaker associations. These discrepant findings may be reflective of small samples and heterogeneity in operationalizing SDOHs and in analytic designs.

Substantial heterogeneity in study methods and statistical approaches precluded a meta-analysis. Although this review supports robust associations between SDOHs and neurobiological alterations, future studies must account for bidirectional processes. SDOH exposure often occurs in early childhood and may exacerbate brain abnormalities or disrupt neurodevelopmental trajectories, while illness-related neurobiological changes may, in turn, contribute to worsening social circumstances. Thus, gaps remain in mapping how structural disadvantage becomes biologically embedded in SSPC.

Future Directions

Research should prioritize integration of multilevel data to identify active components of each SDOH and to clarify pathways connecting environmental exposures to clinical outcomes. Such integration will move the field beyond correlational findings and toward mechanistic models of disease progression. Key future directions are outlined in **Figure 5**.

Although Jeste et al[16] offer a roadmap from social exposures to clinical outcomes via intermediate biological mechanisms, few studies have tested the full pathway or bidirectional effects. Additionally, studies investigating whether pathophysiological processes, like social epigenetic modifications, allostatic load, and inflammaging, mediate the relationship between SDOHs and neurobiological abnormalities are needed.[16] These pathophysiomechanisms could not be evaluated in this review because few studies directly examined them in relation to brain outcomes. Without clear delineation of these processes, the field lacks critical insight into how specific SDOHs impact the brain and how such processes can be therapeutically targeted. Future work should prioritize standardized definitions of exposures and consistent measurement of biological outcomes to facilitate replication and cross-study comparison.[25]

Strengths and Limitations

Certain SDOHs, including food insecurity and racism/discrimination, were infrequently studied in relation to brain outcomes. Moreover, studies rarely assessed intersectionality across different SDOHs. While structural brain changes have been examined extensively, neurochemical and neuroplasticity domains remain underrepresented. Encompassing a broader range of social exposures and neurobiological mechanisms is essential. Studies designed to concurrently capture social, biological, and clinical data can provide comprehensive insight and reveal intersectional effects,[57] such as how multiple SDOHs interact to compound vulnerability or confer resilience.

Timing of SDOH effects is complicated in that studies looked at broad timeframes (ie, childhood and lifetime), and participant age ranged widely between 12 and 65 years. Longitudinal designs are critical for tracking neurodevelopmental timing and trajectories. Adolescence has been identified as a sensitive period for trauma exposure,[6] characterized by ongoing neurodevelopmental changes[58] and heightened HPA axis reactivity and stress-related anxiety responses.[59,60] Evidence further suggests that early-life stress can accelerate neurodevelopmental aging,[61-64] with accelerated aging observed in individuals with or at risk of a SSPC,[65,66] suggesting a biomarker of environmentally mediated disease risk. Longitudinal approaches could illuminate timing and cumulative exposure burden and identify periods of heightened neurobiological vulnerability or resilience. Identifying sensitive windows can help pinpoint when interventions are most effective and inform strategies tailored to developmental stage/illness phase. Enrolling individuals at high risk for psychosis will be informative for linking environmental exposures to conversion risk.[55,67]

Research should move beyond deficit-focused models toward strengths-based frameworks that examine protective factors that confer neurobiological resilience.[5] Factors, such as educational access, strong social networks, financial stability, and equitable health care, may buffer against structural decline, decreased connectivity, or inflammatory dysregulation. Although limited, emerging findings suggest that resilience-promoting environments (eg, neighborhood cohesion)[41] attenuate illness trajectories in at-risk populations. Relatedly, research in other clinical populations has demonstrated the protective effect of trauma-informed therapies in reversing neurobiological alterations associated with trauma.[68,69] Elucidating biological pathways by which protective factors exert their effects could inform strengths-based preventive strategies (eg, family-focused therapy)[70,71] and highlight culturally and contextually relevant sources of resilience.

Clarifying biological pathways through which SDOHs affect the brain is essential for clinical translation. SDOHs are among the most modifiable determinants of mental health.[2] Linking specific exposures to actionable biological processes, such as HPA axis dysfunction, disrupted neural connectivity, or altered neurochemical signaling, can enable targeted interventions.[68,69] Emerging pharmacologic approaches that modulate HPA axis activity (ie, glucocorticoid and vasopressin receptor antagonists, nonpsychoactive cannabinoid receptor agents) are currently under investigation in schizophrenia.[72,73] In parallel, mindfulness-based and stress-management interventions in other clinical populations have demonstrated measurable effects on cortisol awakening response and stress regulation.[74,75] Together, these approaches underscore the translational potential of mechanistically grounded SDOH research.

Conclusions

This systematic review provides critical evidence linking SDOHs with neurobiological abnormalities in SSPC. By tracing pathways through which social determinants may get under the skin, this review underscores the need for equity-informed, biologically grounded approaches to care in SSPCs. Aligning neuroscience with social epidemiology offers the potential to advance prevention, improve treatment, and address the structural conditions that drive mental health disparities.

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Concept and design: All authors.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: All authors.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: All authors.
Administrative, technical, or material support: Hua.
Supervision: All authors.

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